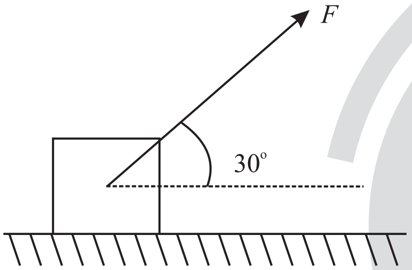
Friction

Single Correct Type Questions

1. As shown in the figure a block of mass 10 *kg* lying on a horizontal surface is pulled by a force F acting at an angle 30^o, with horizontal. For $\mu_s = 0.25$, the block will just start to move for the value of $\vec{F}$: [Given g = $10ms^{-2}$]

[1 Feb, 2023 (Shift-II)]



(*a*) 33.3 *N* (*b*) 25.2 *N*
(*c*) 20 *N* (*d*) 35.7 *N*

2. A body of mass 10 kg is moving with an initial speed of 20 m/s. The body stops after 5 s due to friction between body and the floor. The value of the coefficient of friction is: (Take acceleration due to gravity $g = 10$ ms^{-2})

[31 Jan, 2023 (Shift-II)]

(*a*) 0.2 (*b*) 0.3
(*c*) 0.5 (*d*) 0.4

3. A block of mass 5 kg is placed at rest on a table of rough surface. Now, if a force of 30 N is applied in the direction parallel to surface of the table, the block slides through a distance of 50 m in an interval of time 10 *s*. Coefficient of kinetic friction is (given, $g = 10$ ms^{-2}):

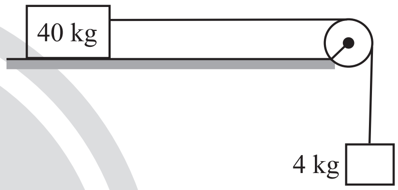
[1 Feb, 2023 (Shift-1)]

(*a*) 0.60 (*b*) 0.75
(*c*) 0.50 (*d*) 0.25

4. A bag is gently dropped on a conveyor belt moving at a speed of 2 m/s. The coefficient of friction between the conveyor belt and bag is 0.4. Initially, the bag slips on the belt before it stops due to friction. The distance travelled by the bag on the belt during slipping motion is:
[Take g = 10 m/s^{-2}] **[27 July, 2022 (Shift-I)]**

(*a*) 2 m (*b*) 0.5 m
(*c*) 3.2 m (*d*) 0.8 m

5. A block of mass 40 kg slides over a surface, when a mass of 4 kg is suspended through an inextensible massless string passing over frictionless pulley as shown below.

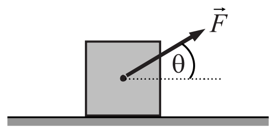


The coefficient of kinetic friction between the surface and block is 0.02. The acceleration of block is.
(given g = 10 ms^{-2}) **[29 June, 2022 (Shift-II)]**

(*a*) 1 ms^{-2} (*b*) 1/5 ms^{-2}
(*c*) 4/5 ms^{-2} (*d*) 8/11 ms^{-2}

6. A block of mass m slides along a floor while a force of magnitude F is applied to it at an angle θ as shown in figure. The coefficient of kinetic is μ_K. Then, the block's acceleration 'a' is given by: **[16 March, 2021 (Shift-I)]**

(g is acceleration due to gravity)



(*a*) $\frac{F}{m}\cos\theta + \mu_K\left(g - \frac{F}{m}\sin\theta\right)$

(*b*) $\frac{F}{m}\cos\theta + \mu_K\left(g + \frac{F}{m}\sin\theta\right)$

(*c*) $\frac{F}{m}\cos\theta - \mu_K\left(g - \frac{F}{m}\sin\theta\right)$

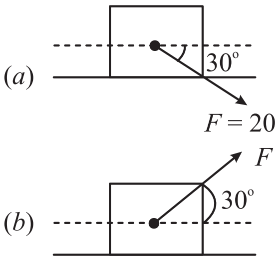
(*d*) $-\frac{F}{m}\cos\theta - \mu_K\left(g - \frac{F}{m}\sin\theta\right)$

7. An insect is at the bottom of a hemispherical ditch of radius 1 m. It crawls up the ditch but starts slipping after it is at height h from the bottom. If the coefficient of friction between the ground and the insect is 0.75, then h is ($g = 10\ ms^{-2}$) **[6 Sep, 2020 (Shift-I)]**

(*a*) 0.45 m (*b*) 0.80 m (*c*) 0.20 m (*d*) 0.60 m

8. A block of mass 5 kg is (i) pushed in case (*a*) and (ii) pulled in case (*b*). by a force $F = 20$ N. Making an angle of 30° with the horizontal, as shown in the figures. The coefficient of friction between the block and floor is $m = 0.2$. The difference between the accelerations of the block, in case (*b*) and case (*a*) will be: ($g = 10\ ms^{-2}$)

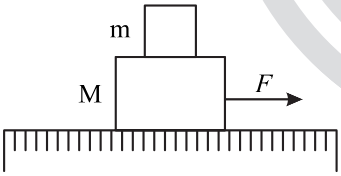
[12 April, 2019 (Shift-II)]



(*a*) $0\ ms^{-2}$ (*b*) $0.8\ ms^{-2}$ (*c*) $0.4\ ms^{-2}$ (*d*) $3.2\ ms^{-2}$

9. A system of two blocks of masses $m = 2$ kg and $m = 8$ kg is placed on a smooth table as shown in figure. The coefficient of static friction between two blocks is 0.5. The maximum horizontal force F that can be applied to the block of mass M so that the blocks move together will be:

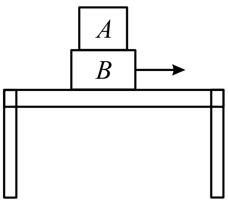
[27 June, 2022 (Shift-I)]



(*a*) 9.8 N (*b*) 39.2 N (*c*) 49 N (*d*) 78.4 N

10. Two blocks A and B of masses $m_A = 1$ kg and $m_B = 3$ kg are kept on the table as shown in figure. The coefficient of friction between A and B is 0.2 and between B and the surface of the table is also 0.2. The maximum force F that can be applied on horizontally, so that the block A does not slide over the block B is: (Take $g = 10\ m/s^2$)

[10 April, 2019 (Shift-II)]



(*a*) 16 N (*b*) 40 N (*c*) 12 N (*d*) 8 N

11. The time taken by an object to slide down a 45° rough inclined plane is n times as it takes to slide down a perfectly smooth 45° inclined plane. The coefficient of kinetic friction between the object and the incline plane is

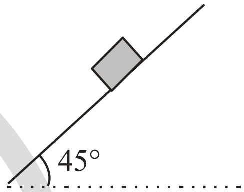
[29 Jan, 2023 (Shift-II)]

(*a*) $\sqrt{\dfrac{1}{1-n^2}}$ (*b*) $\sqrt{1-\dfrac{1}{n^2}}$

(*c*) $1+\dfrac{1}{n^2}$ (*d*) $1-\dfrac{1}{n^2}$

12. Consider a block kept on an inclined plane (inclined at 45°) as shown in the figure. If the force required to just push it up the incline is 2 times the force required to just prevent it from sliding down, the coefficient of friction between the block and inclined plane (μ) is equal to:

[25 Jan, 2023 (Shift-II)]



(*a*) 0.33 (*b*) 0.60

(*c*) 0.25 (*d*) 0.50

13. A block of mass m slides down an inclined plane inclined at angle 30° with an acceleration $\dfrac{g}{4}$. The value of coefficient of kinetic friction will be:

[29 Jan, 2023 (Shift-I)]

(*a*) $\dfrac{2\sqrt{3}+1}{2}$ (*b*) $\dfrac{1}{2\sqrt{3}}$

(*c*) $\dfrac{\sqrt{3}}{2}$ (*d*) $\dfrac{2\sqrt{3}-1}{2}$

14. A block of mass M slides down on a rough inclined plane with constant velocity. The angle made by the incline plane with horizontal is θ. The magnitude of the contact force will be: **[27 July, 2022 (Shift-II)]**

(*a*) Mg

(*b*) $Mg\cos\theta$

(*c*) $\sqrt{Mg\sin\theta + Mg\cos\theta}$

(*d*) $Mg\sin\theta\sqrt{1+\mu}$

15. A block 'A' takes 2 s to slide down a frictionless incline of 30° and length 'ℓ', kept inside a lift going up with uniform velocity 'v'. If the incline is changed to 45°, the time taken by the block, to slide down the incline, will be approximately: **[27 July, 2022 (Shift-II)]**

(*a*) 2.66 s (*b*) 0.83 s

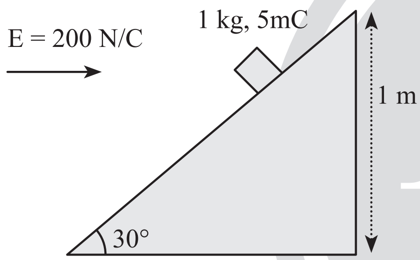
(*c*) 1.68 s (*d*) 0.70 s

16. A block of mass 10 kg, starts sliding on a surface with an initial velocity of 9.8 ms^{-1}. The coefficient of friction between the surface and block is 0.5. The distance covered by the block before coming to rest is:
[Used $g = 9.8$ ms^{-2}] **[24 June, 2022 (Shift-I)]**

(*a*) 4.9 m (*b*) 9.8 m
(*c*) 12.5 m (*d*) 19.6 m

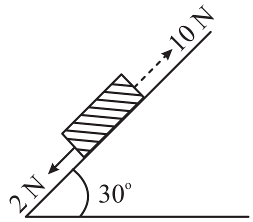
17. An inclined plane making an angle of 30° with the horizontal is placed in a uniform horizontal electric field $200\frac{N}{C}$ as shown in the figure. A body of mass 1 kg and charge 5mC is allowed to slide down from rest at a height of 1m. If the coefficient of friction is 0.2 , find the time taken by the body to reach the bottom.
[26 Feb, 2021 (Shift-II)]

$$\left[g = 9.8\,m/s^2; \sin 30^\circ = \frac{1}{2}; \cos 30^\circ = \frac{\sqrt{3}}{2}\right]$$



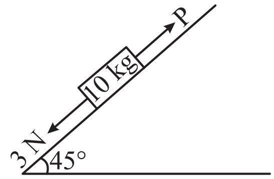
(*a*) 1.3 s (*b*) 0.46 s
(*c*) 2.3 s (*d*) 0.92 s

18. A block kept on a rough inclined plane, as shown in the figure, remains at rest up-to a maximum force 2 N down the inclined plane. The maximum external force up the inclined plane that does not move the block is 10 N. The coefficient of static friction between the block and the plane is:
[12 Jan, 2019 (Shift-II)]



(*a*) $\frac{\sqrt{3}}{2}$ (*b*) $\frac{\sqrt{3}}{4}$
(*c*) $\frac{1}{2}$ (*d*) $\frac{2}{3}$

19. A block of mass 10 kg is kept on a rough inclined plane as shown in the figure. A force of 3 N is applied on the block. The coefficient of static friction between the plane and the block is 0.6. What should be the minimum value of force P, such that the block does not move downward? (Take g = 10 ms^{-2}) **[9 Jan, 2019 (Shift-I)]**



(*a*) 32 N (*b*) 18 N
(*c*) 23 N (*d*) 25 N

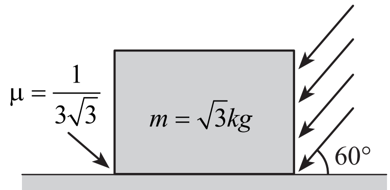
Integer Type Questions

20. A uniform chain of 6 m length is placed on a table such that a part of its length is hanging over the edge of the table. The system is at rest. The co-efficient of static friction between the chain and the surface of the table is 0.5, the maximum length of the chain hanging from the table is _______m. **[25 June, 2022 (Shift-I)]**

21. A body of mass 1 kg rests on a horizontal floor with which it has a coefficient of static friction $\frac{1}{\sqrt{3}}$. It is desired to make the body move by applying the minimum possible force *F* (in N). The value of *F* will be ______ (Round off to the Nearest Integer) [Take $g = 10$ ms^{-2}]
[17 March, 2021 (Shift-II)]

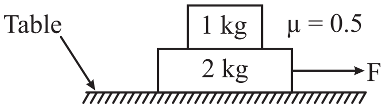
22. The coefficient of static friction between a wooden block of mass 0.5 kg and a vertical rough wall is 0.2. The magnitude of horizontal force that should be applied on the block to keep it adhere to the wall be __________N. [$g = 10$ ms^{-2}] **[24 Feb, 2021 (Shift-I)]**

23. As shown in the figure, a block of mass $\sqrt{3}\,kg$ is kept on a horizontal rough surface of coefficient of friction $\frac{1}{3\sqrt{3}}$. The critical force to be applied on the vertical surface as shown at an angle 60° with horizontal such that it does not move, will be 3*x*. The value of *x* will be
[26 Feb, 2021 (Shift-I)]



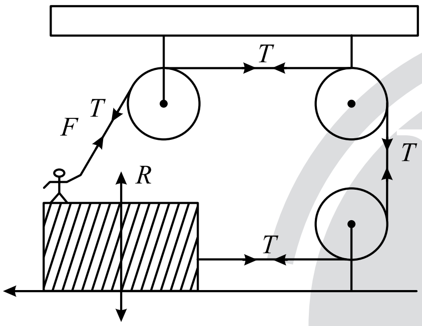
$$\left[g = 10\,m/s^2; \sin 60^\circ = \frac{\sqrt{3}}{2}; \cos 60^\circ = \frac{1}{2}\right]$$

24. The coefficient of static friction between two blocks is 0.5 and the table is smooth. The maximum horizontal force that can be applied to move the blocks together is _______ *N*. (take g = 10 ms^{-2}) **[26 Aug, 2021 (Shift-II)]**

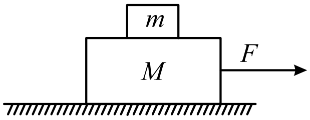


25. A boy of mass 4 kg is standing on a piece of wood having mass 5 kg. If the coefficient of friction between the wood and the floor is 0.5, what is the maximum force that the boy can exert on the rope so that the piece of wood does not move from its place is ______ (in N). (Round off to the Nearest Integer) [Take $g = 10$ ms^{-2}]

[17 March, 2021 (Shift-II)]



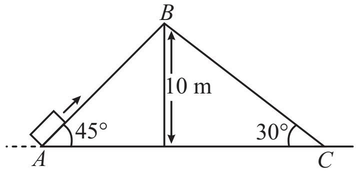
26. Two blocks (m = 0.5 kg and M = 4.5 kg) are arranged on a horizontal frictionless table as shown in figure. The coefficient of static friction between the two blocks is $\frac{3}{7}$. Then the maximum horizontal force that can be applied on the larger block so that the blocks move together is ______ N. (Round off to the Nearest Integer) [Take g as 9.8 ms^{-2}] **[17 March, 2021 (Shift-I)]**



27. Two inclined planes are placed as shown in figure. A block is projected from the point A of inclined plane AB along its surface with velocity just sufficient to carry it to the top Point B at a height 10 m. After reaching the point B the block slide down on inclined plane BC. Time it takes to reach to the point C from point A is $t\left(\sqrt{2}+1\right)s$.

The value of t is _____ (use g = 10 m/s^2)

[27 July, 2022 (Shift-II)]



28. When a body slides down from rest along a smooth inclined plane making an angle of 30° with the horizontal, it takes time T. When the same body slides down from the rest along a rough inclined plane making the same angle and through the same distance, it takes time αT, where α is a constant greater than 1. The co-efficient of friction between the body and the rough plane is $\frac{1}{\sqrt{x}}\left(\frac{\alpha^2-1}{\alpha^2}\right)$ where x = _________.

[1 Sep, 2021 (Shift-II)]

29. A body of mass 'm' is launched up on a rough inclined plane making an angle of 30º with the horizontal. The coefficient of friction between the body and plane is $\frac{\sqrt{x}}{5}$ if the time of ascent is half of the time of descent. The value of x is __________. **[20 July, 2021 (Shift-II)]**

30. A block starts moving up an inclined plane of inclination 30° with an initial velocity of v_0. It comes back to its initial position with velocity $\frac{v_0}{2}$. The value of the coefficient of kinetic friction between the block and the inclined plane is close to $\frac{I}{1000}$. The nearest integer to I is ___________.

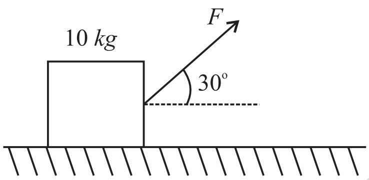
[3 Sep, 2020 (Shift-II)]

ANSWER KEY

1. (*b*)	**2.** (*d*)	**3.** (*c*)	**4.** (*b*)	**5.** (*d*)	**6.** (*c*)	**7.** (*c*)	**8.** (*b*)	**9.** (*c*)	**10.** (*a*)
11. (*d*)	**12.** (*a*)	**13.** (*b*)	**14.** (*a*)	**15.** (*c*)	**16.** (*b*)	**17.** (*a*)	**18.** (*a*)	**19.** (*a*)	**20.** [2]
21. [5]	**22.** [25]	**23.** [3.3]	**24.** [15]	**25.** [30]	**26.** [21]	**27.** [2]	**28.** [$\sqrt{3}$]	**29.** [3]	**30.** [346]

EXPLANATIONS

1. (*b*)



Equating the vertical forces,

$$N = Mg - F\sin 30^\circ = mg - \frac{F}{2} = \frac{200 - F}{2}$$

$$\because F\cos 30^\circ = \mu N$$

$$\sqrt{3}\frac{F}{2} = 0.25 \times \left(\frac{200 - F}{2}\right)$$

$$\Rightarrow F = \frac{200}{4\sqrt{3} + 1} = 25.22N$$

2. (*d*) Using formula, $v = u + at$

$\Rightarrow 0 = 20 + (-\mu g)(5)$ $\quad [\because v = 0,\ a = -\mu g]$

$\Rightarrow \mu = 0.4$

3. (*c*) $S = ut + \frac{1}{2}at^2$

$\therefore\ 50 = 0 + \frac{1}{2} \times a \times 100$

$\Rightarrow a = 1$ m/s^2

$F - \mu mg = ma$

$30 - \mu \times 50 = 5 \times 1$

$\Rightarrow\ m = \frac{1}{2}$

4. (*b*) In frame of belt

$a = \mu g = 4$ m/s^2, $v = 2$m/s, $u = 0$ m/s

$v^2 = u^2 + 2as$

$\Rightarrow s = 0.5$ m

5. (*d*) For 4 kg block

$24g - T = 4a$ $\quad$ ...(*i*)

For 40 kg block

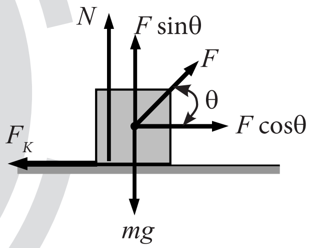
$T - 40g \times 0.02 = 40\,a$ $\quad$ ...(*ii*)

Adding (*i*) and (*ii*)

$40 - 8 = 44a$

$$a = \frac{32}{44} = \frac{8}{11}\,\text{m/s}^2$$

6. (*c*) From the free body diagram given below,



$N = mg - F\sin\theta$ $\quad$ ...(*i*)

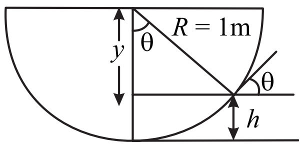
$ma = F\cos\theta - \mu_K N$ $\quad$ ...(*ii*)

On solving (*i*) and (*ii*) we get

$ma = F\cos\theta - \mu_K(mg - F\sin\theta)$

$$\Rightarrow a = \frac{F}{m}\cos\theta - \mu_K\left(g - \frac{F}{m}\sin\theta\right)$$

7. (*c*)



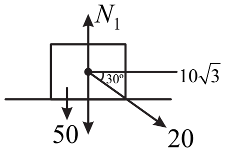
$\mu = \tan\theta$

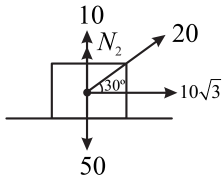
$\Rightarrow \frac{3}{4} = \tan\theta$

$\Rightarrow \theta = 37^\circ$

$\therefore h = R - R\cos\theta = 1 - 1 \times \frac{4}{5} = 0.2$ m

8. (*b*)



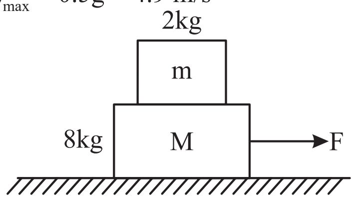


$N_1 = 60,\ N_2 = 40$

$$a_1 = \frac{10\sqrt{3} - 0.2\times 60}{5}$$

$$a_2 = \frac{10\sqrt{3} - 0.2\times 40}{5}$$

9. (*c*) $(a_A)_{max} = 0.5g = 4.9\ m/s^2$



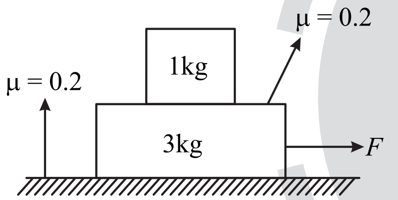
For moving together

$F_{max} = m_T(a_A)$

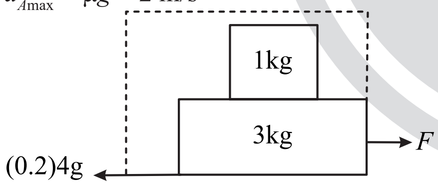
$= 10 \times 4.9$

$= 49$ N

10. (*a*)



$a_{Amax} = \mu g = 2\ m/s^2$



$= 8N$

$F - 8 = 4 \times 2$

$F = 16$ N

11. (*d*) For a perfectly smooth inclined plane

$a_1 = g\sin\theta = g/\sqrt{2}$

For the rough inclined plane

$$a_2 = g\sin\theta - Kg\cos\theta = \frac{g}{\sqrt{2}} - \frac{Kg}{\sqrt{2}}$$

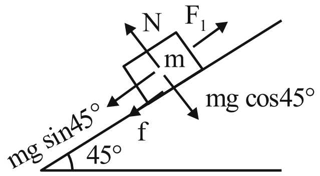
Given, $t_2 = nt_1$

and $a_1t_1^2 = a_2t_2^2$

$$\Rightarrow \frac{g}{\sqrt{2}}t_1^2 = \left(\frac{g}{\sqrt{2}} - \frac{kg}{\sqrt{2}}\right)n^2t_1^2$$

$$\Rightarrow K = 1 - \frac{1}{n^2}$$

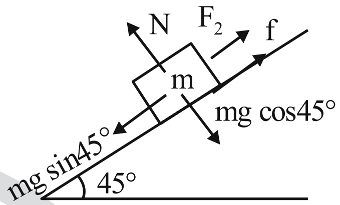
12. (*a*) Force F_1 required to just push the block up the incline



$F_1 = mg\sin 45^o + f$

$$= \frac{mg}{\sqrt{2}} + \mu mg\cos 45^o$$

$$\Rightarrow F_1 = \frac{mg}{\sqrt{2}}(1+\mu)$$



Force F_2 required to prevent the block from sliding

$F_2 = mg\sin 45^o - f$

$= mg\sin 45^o - \mu N$

$$= \frac{mg}{\sqrt{2}}(1-\mu)$$

Given, $F_1 = 2F_2$

$$\Rightarrow \frac{mg}{\sqrt{2}}(1+\mu) = 2\frac{mg}{\sqrt{2}}(1-\mu)$$

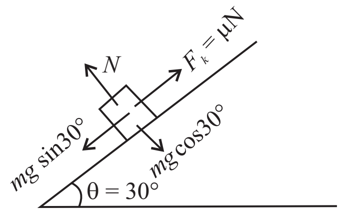
$\Rightarrow \mu = 1/3 = 0.33$

13. (*b*) As the block slides down the plane

$\Rightarrow Mg\sin 30 - f = ma$

$\Rightarrow Mg\sin 30^\circ - \mu mg\cos 30^\circ = ma$

$$\frac{g}{2} - \frac{\sqrt{3}}{2}\cdot\mu g = \frac{g}{4}$$



$$\mu = \frac{1}{2\sqrt{3}}$$

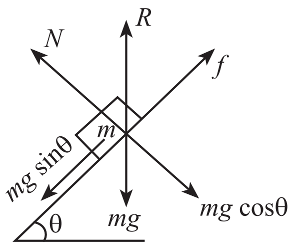
14. (*a*) From F. B. D

$N = Mg\cos\theta$

$f = Mg\sin\theta$

$\therefore R = \sqrt{N^2 + f^2}$

$$= \sqrt{(Mg\cos\theta)^2 + (Mg\sin\theta)^2} = Mg$$



15. (*c*) Acceleration of the block down the inclined plane, $a = g\sin\theta$

From Kinematics equations,

$$\ell = \frac{1}{2} g \sin 30^\circ (2)^2 \quad \text{... } (i)$$

$$\ell = \frac{1}{2} g \sin 45^\circ (t)^2 \quad \text{... } (ii)$$

$(i) = (ii)$

$$\left(\frac{1}{2}\right)(2)^2 = \frac{1}{\sqrt{2}} t^2 \Rightarrow t = \sqrt{2\sqrt{2}} \approx 1.68$$

16. (*b*) $a = -\mu g = -0.5 \times 9.8 = -4.9\ \text{m/s}^2$

$$d = \frac{u^2}{2a} = \frac{9.8 \times 9.8}{2(4.9)}$$

$= 9.8$ m

17. (*a*) Using FBD,

$F = mg\sin\theta - qE\cos\theta - \mu N$

Here $N = mg\cos\theta + qE\sin\theta$

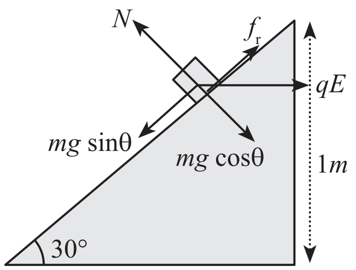
$$\therefore F = \frac{9.8}{2} - \frac{5 \times 10^{-3} \times 200 \times \sqrt{3}}{2}$$

$$-0.2\left(\frac{5 \times 10^{-3} \times 200}{2} + \frac{1 \times 9.8 \times \sqrt{3}}{2}\right)$$

$F = 2.25\ N \Rightarrow a = 2.25\ \text{m/s}^2$

$$S = ut + \frac{1}{2} a t^2$$

$$\Rightarrow t = \sqrt{\frac{2\ell}{a}} = 1.31 \text{sec}$$



18. (*a*) $mg\sin\theta + 2 = \mu\, mg\cos\theta$ ... (*i*)

$10 - mg\sin\theta = \mu\, mg\cos\theta$ ... (*ii*)

On adding (*i*) + (*ii*)

$$12 = 2\,\mu mg\,\frac{\sqrt{3}}{2};\ \mu mg = \frac{12}{\sqrt{3}}$$

On (*a*) – (*b*):

$$8 = 2mg \times \frac{1}{2};\ mg = 8;\ \mu = \frac{\sqrt{3}}{2}$$

19. (*a*) For equilibrium of the block net force be zero. Hence we can write.

$mg\sin\theta + 3 = P + \text{friction}$

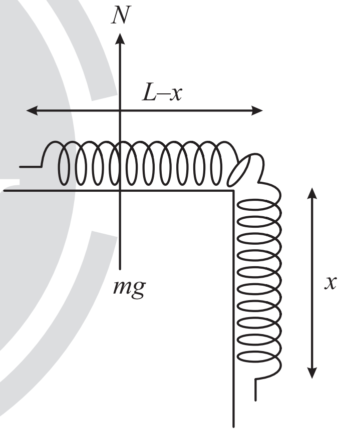
$\text{mg}\sin\theta + 3 = P + \mu mg\cos\theta$

After solving, we get, $P = 32$ N

20. [2] Mass per unit length = λ

$N = mg = \lambda(L - x)g$

$f_{s,\max} = \mu_s N$



$f_{s,\max} = (0.5)\,(\lambda)\,(L - x)g$

And also $f_{s,\max} = m_x g$

$0.5\lambda\,(L - x)g = \lambda x g$

$$\frac{L - x}{2} = x$$

$$\frac{L}{2} = \frac{3x}{2} \Rightarrow x = \frac{L}{3} = \frac{6}{3} = 2\ \text{m}$$

21. [5] Minimum possible force,

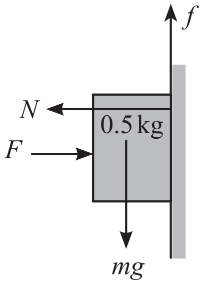
$$F = \frac{\mu mg}{\sqrt{1 + \mu^2}}$$

$$F_{\min} = \frac{\frac{1}{\sqrt{3}} \times 1 \times 10}{\sqrt{1 + \frac{1}{3}}} = 5\text{N}$$

22. [25] $f = W = 0.5 \times 10 = 5\ N$

$N = F$

The magnitude of horizontal force should be smaller man frictional force to keep it adhere to the wall.



$\Rightarrow f \le \mu N$

$\Rightarrow 5 \le 0.2\ F$

$\Rightarrow F \ge 25\ N$

23. [3.3] Net force in the Vertical direction is zero so.

$N = mg + F \sin 60°$

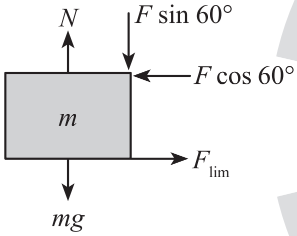
$f_{\ell im} = F\cos 60° \Rightarrow \mu N = F\cos 60°$

$\Rightarrow \mu(mg + F\sin 60°) = F\cos 60°$

$\Rightarrow \mu mg = F[\cos 60° - \mu \sin 60°]$

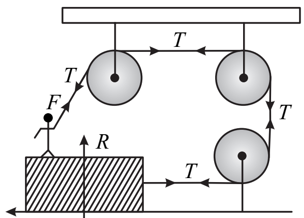
$$F = \frac{\mu mg}{\cos 60^\circ - \mu \sin 60^\circ} = \frac{\frac{1}{3\sqrt{3}} \times \sqrt{3} \times 10}{\frac{1}{2} - \frac{1}{3\sqrt{3}} \times \frac{\sqrt{3}}{2}} = \frac{10/3}{\frac{1}{2} - \frac{1}{6}}$$

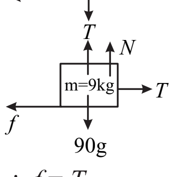
$$= \frac{10/3}{1/3} = 10\ N = 3 \times \left(\frac{10}{3}\right) N$$



24. [15] $\frac{F_{max}}{1+2} = 0.5 \times 10 \Rightarrow F_{max} = 15\ N$

25. [30]





$\therefore f = T$

$\Rightarrow \mu N = T$

$\Rightarrow \mu\ (90 - T) = T$

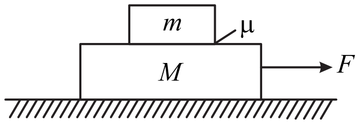
$\Rightarrow 0.5\ (90 - T) = T$

$\Rightarrow 90 - T = 2T$

$\Rightarrow 3T = 90$

$\Rightarrow T = 30\ N$

26. [21]



$$a = \frac{F}{M+m}$$

$$f = ma = m\frac{F}{M+m}$$

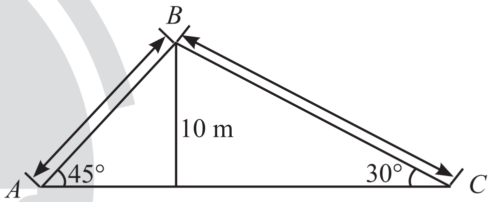
$$m\frac{F}{M+m} \le \mu mg \quad \text{(for no slipping)}$$

$$\Rightarrow F \le \mu(m+M)g$$

$$\therefore F_{max} = \frac{3}{7}(0.5+4.5) \times 9.8 = 21N$$

27. [2] Acceleration between A and $B = -g\sin 45° = \frac{-g}{\sqrt{2}}$

Acceleration between B and $C = +g\sin 30° = \frac{g}{2}$



Velocity at point A.

$$0 = u^2 - 2\frac{g}{\sqrt{2}} \times 10\sqrt{2}$$

$u = 10\sqrt{2}$ m/s

From A to B

$v = u + at$

$$0 = 10\sqrt{2} - \frac{g}{\sqrt{2}} t_{AB}$$

$t_{AB} = 25$ ...(*i*)

From B to C

$$s = ut + \frac{1}{2}at^2$$

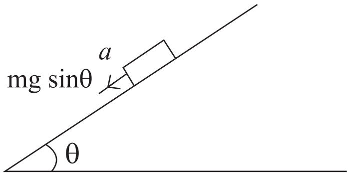
$$20 = 0 + \frac{1}{2} \times \frac{g}{2} t_{BC}^2$$

$t_{BC} = 2\sqrt{2}$ ...(*ii*)

$t_{AB} + t_{BC} = 2\left(\sqrt{2}+1\right)$

$\Rightarrow t = 2$

28. [$\sqrt{3}$] For smooth incline surface

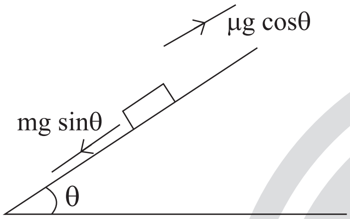


$$a = g\sin 30^\circ = \frac{g}{2}$$

$$S_1 = ut + \frac{1}{2}at^2$$

$$S_1 = \frac{1}{2}\frac{g}{2}t^2 = \frac{g}{4}t^2 \qquad ...(i)$$

For rough incline surface



$a = g\sin\theta - \mu g\cos\theta$

$$a = \left(\frac{g}{2} - \mu g \times \frac{\sqrt{3}}{2}\right) = \frac{g}{2}(1-\mu\sqrt{3})$$

Now

$$S = \frac{1}{2}a(\alpha t)^2$$

$$S = \frac{1}{2}\frac{g}{2}(1-\mu\sqrt{3})\alpha^2 t^2 \qquad ...(ii)$$

Solving (*i*) and (*ii*)

$$\frac{g}{4}t^2 = \frac{g}{4}(1-\mu\sqrt{3})\alpha^2 t^2$$

$$1 = (1-\mu\sqrt{3})\alpha^2$$

$$\mu = \frac{1}{\sqrt{3}}\left(\frac{\alpha^2 - 1}{\alpha^2}\right)$$

$$\Rightarrow x = \sqrt{3}$$

29. [3] For Ascent:

$$t_a = \sqrt{\frac{2l}{a_{ascent}}} = \sqrt{\frac{2l}{g(\sin\theta + \mu\cos\theta)}} \qquad ...(i)$$

For Descent:

$$t_d = \sqrt{\frac{2l}{a_{descent}}} = \sqrt{\frac{2l}{g(\sin\theta - \mu\cos\theta)}} \qquad ...(ii)$$

According to question, we can write

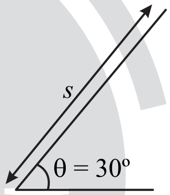
$$t_a = \frac{1}{2}t_d$$

$$\Rightarrow \frac{1}{\sin\theta + \mu\cos\theta} = \left(\frac{1}{4}\right)\left(\frac{1}{\sin\theta - \mu\cos\theta}\right)$$

[From (*i*) and (*ii*)]

$$\Rightarrow \mu = \frac{3}{5}\tan\theta = \frac{3}{5}\tan 30^\circ = \frac{\sqrt{3}}{5}$$

30. [346] For upward motion



$$\frac{1}{2}mv_0^2 = \left(mg\frac{S}{2} + \mu mg\frac{\sqrt{3}}{2}S\right)$$

$$\frac{1}{2}\frac{mv_0^2}{4} = \left(mg\frac{S}{2} - \mu mg\frac{\sqrt{3}}{2}S\right)$$

For downward

$$4 = \frac{(1+\mu\sqrt{3})}{(1-\mu\sqrt{3})}$$

$$\Rightarrow 4 - 4\mu\sqrt{3} = 1 + \mu\sqrt{3}$$

$$\Rightarrow 3 = 5\mu\sqrt{3} \qquad \therefore \mu = \frac{\sqrt{3}}{5} \approx \frac{346.41}{1000}$$

$I = 346$